

RADIAN - RADar-based Intelligent
Activity moNitoring
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CONCEPT OF OPERATIONS

CONCEPT OF OPERATIONS FOR RADIAN

TEAM <60>

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1. Executive Summary

Falls are a leading cause of injury among elderly individuals, often resulting in severe health consequences. Existing monitoring solutions on the market, such as wearable devices and camera-based systems, suffer from significant drawbacks. Wearable devices rely on user compliance, and camera-based systems compromise privacy. To address these drawbacks, this project proposes the development of a low-cost, AI-enabled fall detection system using a millimeter wave (mmWave) radar marketed for the elderly at risk of falls.

The RADIAN system uses a commercially available mmWave radar sensor to continuously monitor human activity within an indoor environment, such as a living room or bedroom. The use of mmWave technology allows for privacy to be preserved while still providing adequate protection. The data from the radar will be processed through an embedded microcontroller and then passed to a Raspberry Pi, where AI algorithms will classify motion patterns to detect falls. In the event of a fall, the RADIAN system will automatically send a wireless alert to a caregiver's smartphone via Wi-Fi, as well as set off an alarm on the device itself to alert anyone inside the house. The budget approved for the RADIAN group is \$375, keeping the project low-cost.

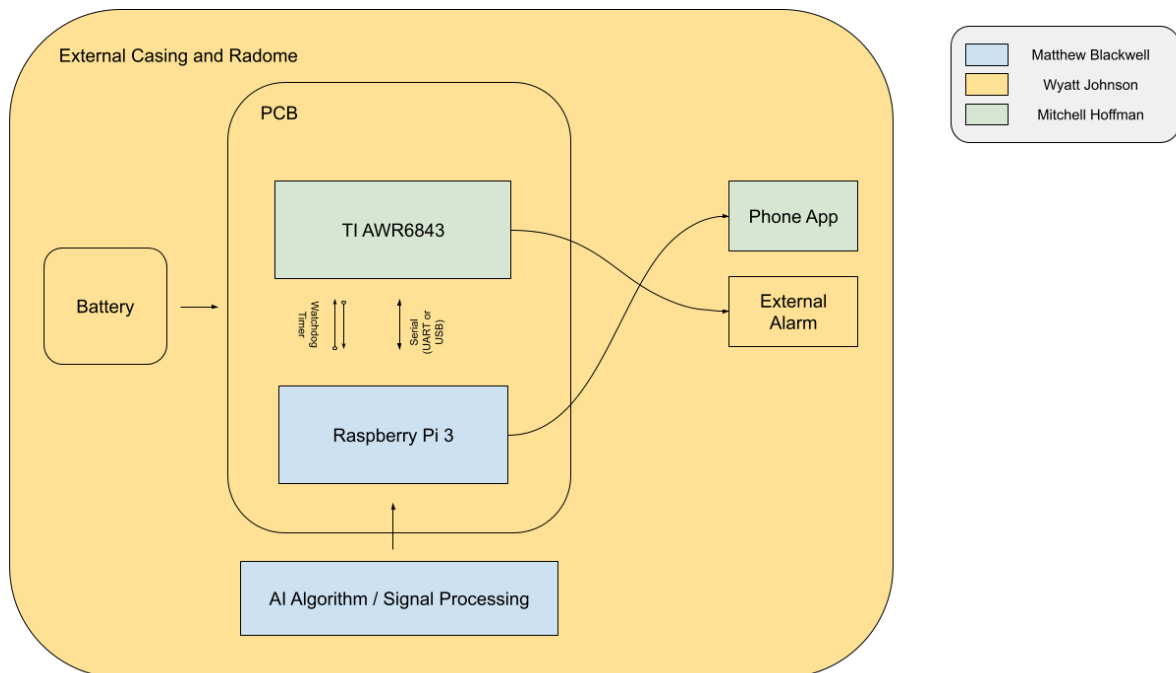
2. Introduction

The RADIAN (RADar-based Intelligent Activity moNitoring) system was created to detect human falls within an enclosed space, specifically in a household or office setting, using radar-based systems. The RADIAN system will detect a fall using AI-based machine learning algorithms and send an alert to a cellular device, ensuring that the fall is caught quickly and accurately.

2.1. Background

The leading cause of injury-related death among adults ages 65 and older is falls [1]. Many current solutions rely on vision-based systems or a wearable device, both of which suffer from limitations such as user compliance, privacy concerns, and coverage gaps. A millimeter-wave radar preserves privacy, a non-contact, and reliable means of monitoring human activity indoors.

2.2. Overview



2.2.1 Figure 1: High-Level Block Diagram for RADIAN system

A mmWave Radar Sensor will be used to monitor human movement in a room. Data from the radar is processed using AI algorithms to detect falls in real time. Once a fall is detected, the device will activate an alarm as well as send a remote wireless notification to a mobile device via Wi-Fi.

2.3. Referenced Documents and Standards

- IEEE 802.11 Wi-Fi standards
 - IEEE 802.15.4 for wireless communications
 - IEEE 521-2002 Standard Letter Designations for Radar-Frequency Bands
- [1] "Falls Data & Research." Centers for Disease Control and Prevention, Accessed at <https://www.cdc.gov/falls/data-research/index.html#:~:text=Deaths%20from%20older%20adult%20falls,.cdc.gov/STEAD>

3. Operating Concept

3.1. Scope

The RADar-based Intelligent Activity moNitoring (RADIAN) system is designed to enhance the safety of elderly and physically disabled individuals by providing timely alerts in the event of a fall within a monitored room. Once installed, the RADIAN system will continually monitor a closed room for human presence and movement. Fall events are identified by an in-house-developed artificial intelligence (AI) and machine learning algorithm, based on extensive testing by the RADIAN group. When the AI determines that a fall has occurred, a wireless signal will transmit to a dedicated mobile application developed by the RADIAN group, alerting those most important to the monitored individual. In addition, there will be an audible alarm system to alert those in the building to the fall. By combining real-time monitoring with reliable alerting mechanisms, the RADIAN system allows for peace of mind for at-risk individuals and their families.

3.2. Operational Description and Constraints

The RADIAN system is designed to be used in closed rooms of varying sizes, both in room area and ceiling height. The RADIAN system is designed to be user-friendly, both in the setup of the radar as well as the alert system and its GUI. A RADIAN system will be set up in the corner of a room, with the sensor and radome facing down at a 45° angle to ensure that the RADIAN system is scanning the entire room. The GUI (Graphical User Interface) will ask what room the RADIAN system is set up in to organize each RADIAN system and to understand what furniture and other objects could be scanned. As the RADIAN system scans the room, the raw data will be processed through the microcontroller embedded in the RADIAN system and then sent to an AI algorithm to analyze the room. If the AI detects a fall, then a signal will be sent to the microcontroller to then alert the GUI that a fall was detected through an app notification.

Here are the resulting constraints of the RADIAN system for it to operate properly according to the description:

- The RADIAN system must not be blocked by any object
- The RADIAN system has to be placed in a high vantage point at a 45° downward angle
- The RADIAN system's radome must be directed towards the largest area of the room
- The RADIAN system must be powered via battery
- The RADIAN system must be small in size (e.g., wall-mountable, no larger than a shoebox) to ensure user-friendliness during installation, thus limiting the complexity of parts that can be used
- The RADIAN group has a budget of \$375, further constraining the complexity of the RADIAN system

3.3. System Description

- **Physical Design:** The radar board, which includes a TI AWR6843 and a Raspberry Pi, will all be connected to the battery system on a custom-designed PCB (Printed Circuit Board) designed in-house by the RADIAN group. This PCB design will enhance the RADIAN system's reliability during installation, as no components can be removed. The RADIAN group will also study and implement the power management in this subsystem, ensuring that the correct voltage and current are being provided to the various components of the RADIAN system. The battery casing will also be soldered to the board and include a removable housing and leads to facilitate battery replacement when needed, ensuring the continued reliability of the RADIAN system.
- **Wireless Design:** The radar chip will be hand-selected through thorough research and trade studies by the RADIAN group to guarantee reliability and affordability for the consumer and users of the RADIAN system. The TI radar board will be programmed to collect the raw data from the radar chip and convert it into usable data for the AI algorithm stored on the Raspberry Pi. Once the AI algorithm has detected a fall, a signal will be sent to an external alarm connected to the RADIAN system. The wireless design subsystem will also be responsible for the application's GUI, providing both a frontend and a backend to ensure that the application is easy to use and runs smoothly.
- **Data Design:** The AI algorithm is the primary focus of the data subsystem, utilizing a simple AI model developed by the RADIAN group to predict fall occurrences. The RADIAN group will do extensive testing and training of the AI model to ensure that false positives and true negatives do not send alerts. The application may have various options to select where the RADIAN system is placed, to help the AI algorithm understand what objects may or may not be in a room. Some examples include a bed present in a bedroom, but not in a living or dining room. Having these distinctions allows for fewer false alarms when the user is lying down vs. falling. The data subsystem will also be responsible for communicating with the cloud server to send the push notifications. A wireless Wi-Fi communication will be sent from the Raspberry Pi 3 to the Gotify cloud to alert those not in proximity of the audible alarm using the RADIAN phone application.

3.4. Modes of Operations

The RADIAN system will only have one operating mode, as the radar has to constantly sweep a room due to the random nature of human movements. Within this mode, the user will be able to define what room the RADIAN system is placed in to aid the AI algorithm in understanding what may or may not be in the room. When operating in the mode, the radar will constantly detect the room and the objects around the room, sensing for motion and a potential fall. If a fall is detected by the

RADIAN system and its AI algorithm, then an alert will be sent to the desired device as well as set off an audible alarm for those in proximity.

3.5. Users

The RADIAN system will be marketed to the high-risk individuals who would like a fall detection system in their home without having to worry about privacy infringement with camera-based detection systems. Our product will allow users and their loved ones to have peace of mind when it comes to the health and safety of the user.

The RADIAN system can also double as a simple motion detection device that can be used in various security applications, including home security, business security, and preventing deaths in the back of motor vehicles.

3.6. Support

The RADIAN group will provide an extensive manual detailing the installation of the RADIAN system, battery replacement, and a simple explanation of how the RADIAN system works at the highest level. Within the RADIAN application, detailed step-by-step guides will be provided to guide the user through setup and operation of the device.

4. Scenario(s)

4.1. Fall Detection

An elderly person falls in their bedroom or living room. The RADIAN system will use mmWave radar technology to detect these sudden movements. This mmWave radar data will be sent to an AI model that will then classify the event as a fall, triggering an alarm and notifying the caretaker. This notification will be sent to a caregiver's smartphone through an app. The caregiver will then be able to check on the individual quickly and allow for the necessary response as soon as possible.

4.2. Normal Activity Monitoring

The RADIAN system user can be walking, sitting, or lying down in their bedroom. The radar will detect all of these events, but the AI model will classify these events as "safe". No alerts will be sent in these events, and will allow the user to continuously monitor their environment for hazardous falls without invading their privacy.

4.3. Multi-Room Monitoring

The user is capable of monitoring a whole house or a nursing home. The user will install multiple RADIAN systems in each room that they would like to monitor. Each RADIAN system will individually monitor the room in which it is. If the RADIAN system detects a fall, it will notify the caregiver of the event and also provide the location of the room in which it occurred. While one room is triggered, the other rooms will be continuously monitored. All of the data from the rooms can be examined in the RADIAN smartphone app by the caregiver, providing complete protection of a whole building.

5. Analysis

5.1. Summary of Proposed Improvements

- The RADIAN system will be battery-powered, providing wireless capabilities and flexible placement without requiring a constant wired connection.
- The radar subsystem will monitor human activity without capturing images, which allows for the protection of the privacy of users while still detecting falls.
- The AI model constantly monitors the radar subsystem's data to detect falls and then sends notifications.
- Alerts will be sent directly to the caregivers or emergency responders, allowing for a quick response time to accidents.
- The RADIAN system will operate autonomously, allowing users to have minimal interaction with the device.

5.2. Disadvantages and Limitations

- Due to the limited power, since it will be battery-powered, this can restrict our complexity for our AI algorithm. This could lead to false alarms and/or missing scenarios that should trigger the alarm.
- Depending on the sampling rate and wireless communication frequency, this can cause the battery life to be significantly reduced.
- In complex environments (cluttered rooms, multiple people, animals, children) the RADIAN system could produce more false positives.
- Depending on the size of the room, the RADIAN systems may not be able to detect falls in the outer areas of a large room.
- For the RADIAN system to effectively communicate with emergency services, it will need to depend on reliable network connectivity.

5.3. Alternatives

PIR Motion Sensors

- Used for simple movement/presence detection.
- Cheap and easy to use.
- Can not detect falls.

Camera-Based Monitoring

- Can provide a visual view of the room and verify if a fall occurs.
- Offers more data for an AI model and could be used to create a better model without the need for complex data or high-resolution cameras.
- Raises privacy concerns for bedrooms and living spaces.
- Requires high processing power, and storage could be an issue, leading to a high price point.

Wearable device

- Capable of detecting a fall with an onboard sensor and more accurate, with little complexity.
- Requires the user to wear a device constantly.

- It may be uncomfortable, or the user could forget to wear the device, making it not as reliable

5.4. Impact

Environmental Impact

- The RADIAN device has low energy consumption.
- Minimal waste since the device will be compacted.

Society Impact

- Increases safety for the elderly by providing rapid fall detection without invading privacy
- Reduces the response times for caregivers and/or emergency services
- Enables the elderly to live independently without sacrificing safety

Ethical concerns

- Allows for privacy due to no audio or video recording.

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FUNCTIONAL SYSTEM REQUIREMENTS

FUNCTIONAL SYSTEM REQUIREMENTS FOR RADIAN

PREPARED BY:

Author Date

APPROVED BY:

Matthew Blackwell Date

Prof. Jang Date

Roman Venegas Date

Change Record

Rev	Date	Originator	Approvals	Description
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1. Introduction

1.1. Purpose and Scope

Falls are a leading cause of injury among elderly individuals, often resulting in severe health consequences. Existing monitoring solutions on the market, such as wearable devices and camera-based systems, suffer from significant drawbacks. Wearable devices rely on user compliance, and camera-based systems compromise privacy. The RADIAN system will provide a low-cost, non-invasive, privacy-preserving fall detection system. The RADIAN system will use mmWave radar and AI algorithm technology to monitor human activity and detect falls in indoor environments, such as bedrooms and living rooms.

The RADIAN system will continuously monitor a room for motion patterns and classify them into fall/non-fall events in real time. Once a fall is detected, the RADIAN system will trigger an audible alarm and send a wireless alert to a caregiver's smartphone. Figure 1 shows the RADIAN system that is proposed in CONOPS.

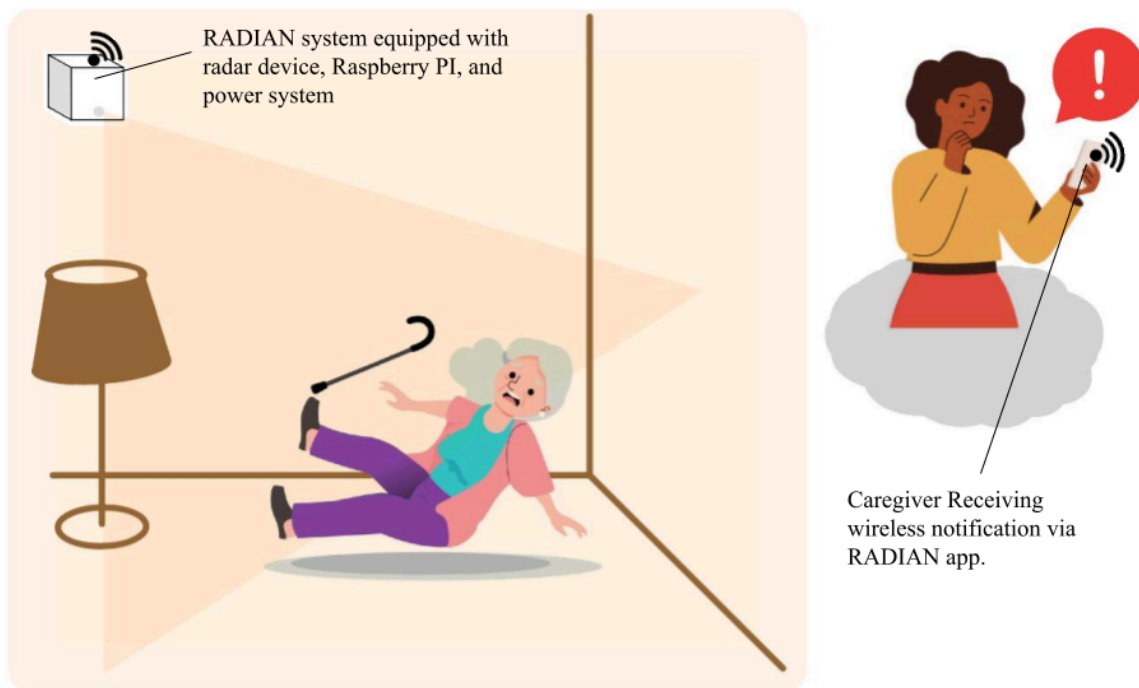


Figure 1. RADIAN Conceptual Image

This document describes detailed requirements for the RADIAN system, including the requirements for each subsystem (Physical, Wireless, and Data). The verification process for the RADIAN system will be described in a separate Validation Plan.

1.2. Responsibility and Change Authority

Matthew Blackwell, RADIAN's team leader, will be responsible for ensuring that all requirements are met for the RADIAN system. These requirements can only be changed with approval from Matthew Blackwell and Professor Arya Menon, the RADIAN faculty Contact.

Subsystem	Responsibility
AI Model and Communication	Matthew Blackwell
Radar Sensing and App Development	Mitchell Hoffman
Power System and Physical Enclosure	Wyatt Johnson

Table 1: Subsystem responsibility

2. Applicable and Reference Documents

2.1. Applicable Documents

The following documents, of the exact issue and revision shown, form a part of this specification to the extent specified herein:

Document Number	Revision/Release Date	Document Title
IPC/WHMA-A-620E	January 2022	Requirements and Acceptance for Cable and Wire Harness Assemblies
UL 1977	December 7, 2022	Component Connectors for Use in Data, Signal, Control and Power Applications

Table 2: Applicable Documents for the RADIAN System

2.2. Reference Documents

The following documents are reference documents utilized in the development of this specification. These documents do not form a part of this specification and are not controlled by their reference herein.

Document Number	Revision/Release Date	Document Title
IEEE 802.11	Jan. 23, 1996	IEEE Standard for Information Technology – Telecommunications and Information Exchange between systems

Table 3: Reference Documents for the RADIAN System

2.3. *Order of Precedence*

In the event of a conflict between the text of this specification and an applicable document cited herein, the text of this specification takes precedence without any exceptions.

All specifications, standards, exhibits, drawings, or other documents that are invoked as “applicable” in this specification are incorporated as cited. All documents that are referred to within an applicable report are considered to be for guidance and information only, except ICD,s which have their relevant documents considered to be incorporated as cited.

3. Requirements

3.1. System Definition

The RADIAN system is an all-in-one, designed system to help protect the elderly and high-risk individuals with quick and accurate alerts when the user has fallen. This allows the caregiver, whether a health care professional or loved one, to feel comfort knowing that their loved one is safe. The RADIAN system will comprise three overarching subsystems, each with further design requirements that must be met for the RADIAN subsystem. These three subsystems include Physical, Data, and Wireless.

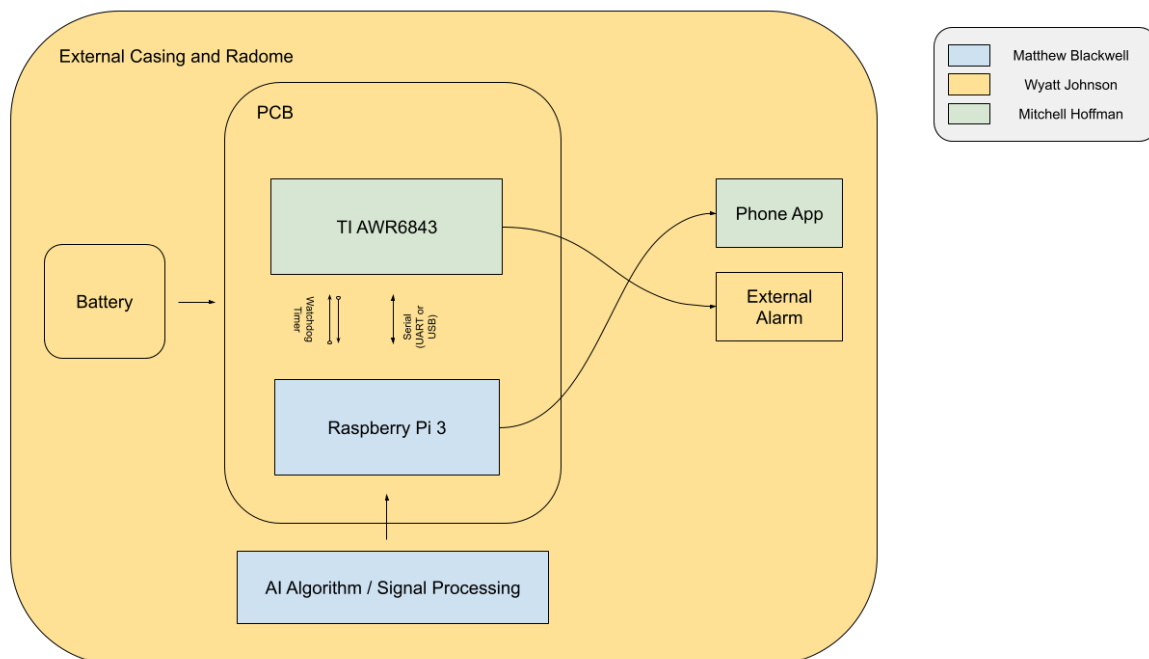


Figure 2. Block Diagram of Radian System

The entire RADIAN system is powered through two 3.7V rechargeable batteries, powering both the TI AWR6843ISK board and the Raspberry Pi 3, which are needed for the system to function. The rechargeable battery pack allows the user to easily charge the RADIAN system while maintaining maneuverability around the room. Both the TI board and Raspberry Pi 3 are connected to the custom-designed PCB, allowing for efficient connections and reduced risk of electrical connection failure between the subsystems. The ML model and AI algorithm are stored and run through the Raspberry Pi 3, where they will then communicate with an external phone application designed by the RADIAN group. An external alarm will also be connected to the RADIAN system to alert anyone in the home while also alerting the caregiver's device.

The RADIAN system will have a feature that will trigger a sleep mode on the Raspberry Pi 3 to conserve power when no motion is detected in the room. Once motion is detected of an elderly or high-risk individual, the radar will continuously scan the room, sending the processed data to the Raspberry Pi 3. Through its AI algorithm and ML model, the Raspberry Pi 3 will be able to compute in real-time the position, velocity, acceleration, and other fall parameters of the individual. If the ML model decides that a true fall has occurred based on a variety of parameters and decisions, the Raspberry Pi will send a message. The Gotify server will receive this new message and create a WebSocket, allowing for the message to travel in real time to the phone application. The custom-designed application will then receive this message and send a push notification to the caregiver, alerting them of the fall.

By combining these three subsystems, RADIAN provides a portable, intelligent, and connected solution for fall detection, balancing energy efficiency with reliable caregiver communication.

3.2. Characteristics

3.2.1. Functional / Performance Requirements

3.2.1.1. Fall Detection Accuracy

The RADIAN system shall detect falls with a minimum of 90% accuracy under normal indoor conditions (room temperature, clear view of the falling subject, unobstructed view, within 5 meters).

Rationale: The system will need to have a high fall detection accuracy to verify its capabilities for detecting falls reliably, which will also increase user trust.

3.2.1.2. Detection latency

Once a fall is detected, the RADIAN system will send an alert and sound an alarm within five seconds of a fall.

Rationale: It is critical that caregivers or emergency responders act as soon as possible, reducing the chance of an injury worsening.

3.2.1.3. Battery Life

The RADIAN system shall operate for a minimum of 4 hours on battery power without recharging.

Rationale: This amount of time will allow the system to stay functioning while the second battery is charging.

3.2.1.4. Communication Reliability

The RADIAN system shall send an alert over the Gotify server with a minimum success rate of 95%.

Rationale: A reliable communication system will ensure that fall events reach caregivers promptly, allowing for fast response times.

3.2.1.5. Local Alert Functionality

The RADIAN system will be equipped with a local audible alert system that, upon a fall being detected, a minimum of a 70 dB alarm will be activated.

Rationale: Having the local alert system will provide an alert to the user and nearby individuals that a fall has occurred. This will also allow for locating the victim quickly.

3.2.1.6. Autonomous Operation

The RADIAN system will operate autonomously, allowing users to have minimal interaction with the device.

Rationale: Autonomy ensures there will be continuous monitoring and timely alerts without the user needing to worry about maintaining the device.

3.2.2. Physical Characteristics

3.2.2.1. Radar Scan Area

The RADIAN system will operate in a standard room of 20 to 30 square meters with an angle of 45 degrees facing down.

Rationale: The RADIAN system is intended to operate in any standard household room, including but not limited to dining rooms, living rooms, and bedrooms. However, the RADIAN system is not rated for the bathroom.

3.2.2.2. Mass

The mass of the RADIAN System shall be less than or equal to 0.8 kilograms.

Rationale: A lighter system will be able to be mounted on walls without requiring heavy-duty hardware and support.

3.2.2.3. Volume Envelope

The volume envelope of the RADIAN System shall be less than or equal to 5 inches in height, 8 inches in width, and 14 inches in length.

Rationale: A compact system will be able to be mounted on walls, ceiling, or furniture, allowing for portability & relocation.

3.2.2.4. Mounting

The mounting information will be available in the RADIAN group's ICD as well as their user guide.

Rationale: The RADIAN system must be placed at a certain angle and height to capture the entirety of the room it is placed in.

3.2.2.5. Installation

The instructions for installing the RADIAN system are available for reference in the provided user guide. The installation required by the user will be mounting and battery charging/installation, as the phone application will already be installed and set up.

Rationale: The ease of installation allows the user to mount the RADIAN system and use it, ensuring that anyone can use the system.

3.2.3. Electrical Characteristics

3.2.3.1. Inputs

- a. The presence or absence of any combination of the input signals in accordance with ICD specifications applied in any sequence shall not damage the RADIAN system, reduce its life expectancy, or cause any malfunction, either when the unit is powered or when it is not.
- b. No sequence of commands shall damage the RADIAN System, reduce its life expectancy, or cause any malfunction.

Rationale: By design, it should limit the chance of damage or malfunction by user/technician error.

3.2.3.1.1 Power Consumption

The maximum peak power of the system shall not exceed 8 watts.

Rationale: Lower power draw will allow the system to run longer, while a high wattage will shorten the runtime.

3.2.3.1.2 Input Voltage Level

The input voltage level for the RADIAN system shall be 5.0V.

Rationale: This is the required voltage for devices used in the RADIAN system

3.2.3.2. Outputs

3.2.3.2.1 Data Output

The RADIAN system shall include a phone application interface compatible with the data system.

Rationale: The RADIAN system will pass information, specifically if a fall was detected, to a phone app designed for the caregiver of the high-risk individual.

3.2.3.2.2 Diagnostic Output

The RADIAN system may include a diagnostic interface for errors that occur within the system and debugging techniques for the system.

Rationale: This will provide a way for the use of the phone app to locate the error and provide solutions for fixing the error.

3.2.3.2.3 Wireless Communication Output

The RADIAN system shall wirelessly communicate with the phone application to alert if a fall is detected, using the Gotify server.

Rationale: Wireless communication will allow the caregiver to be notified, even if they are not present in the home to hear the audible alarm.

3.2.3.2.4 Cloud Data Storage

The RADIAN system may include a cloud storage option to upload data 4 seconds before the fall, during the fall, and 10 seconds after the fall.

Rationale: Cloud storage allows for first responders to know exactly when a fall occurs and the extent of the severity of the fall.

3.2.3.3. Connectors

The RADIAN System shall follow UL 1977 (2022), Component Connectors for Use in Data, Signal, Control, and Power Applications.

Rationale: Following UL 1977 will ensure that we use safe techniques and correctly rated connectors for lower power systems.

3.2.3.4. Wiring

The RADIAN System shall follow the guidelines outlined in IPC/WHMA-A-620 Class 1 safety requirements.

Rationale: Class 1 establishes basic workmanship for general, low-voltage products.

3.2.4. Environmental Requirements

The RADIAN System shall be designed to withstand and operate in the environments and laboratory tests specified in the following section.

Rationale: The RADIAN system will need to function in an indoor setting similar to the common home, with constant pressure and slight variations in temperature

3.2.4.1. Thermal

The RADIAN System will be able to operate in an environment between 50 °F and 80 °F

Rationale: The average temperature of many homes and buildings is between 50 °F and 80 °F, so the system must accommodate these varying temperatures

4. Support Requirements

4.1. Wi-Fi Connection

The RADIAN system requires a Wi-Fi connection to communicate with the Gotify server to send the REST POST notification. The user will provide the Wi-Fi connection to support the wireless connection of the RADIAN system.

Rationale: The system will need a Wi-Fi connection for the Raspberry Pi 3 to communicate with the Gotify cloud server. The Raspberry Pi 3's Wi-Fi connection can be changed remotely without having to view the GUI of the device.

4.2. Wi-Fi or Cellular Data Connection

The caregiver will be required to have a Wi-Fi or Cellular Data connection to receive the push notification from the RADIAN phone application.

Rationale: This allows the push notification to be communicated wirelessly over any distance.

4.3. Battery Maintenance

The user or caregiver must maintain the RADIAN system by occasionally changing the rechargeable batteries. They must also charge the depleted batteries to ensure there is always a backup for the system.

Rationale: Having battery maintenance allows the system to operate without protruding cables and will work if a power outage occurs.

4.4. Requirements for Customer

The RADIAN system requires a Wi-Fi connection to post an alert to the Gotify cloud server properly. Due to the system's nature, the caretaker will also need to download the phone application to receive push notifications indicating a fall has occurred. The user will have to provide the Wi-Fi connection, but this should not be an issue, as the majority of homes and buildings have some sort of Wi-Fi connection. The caretaker must have a wireless connection, either Wi-Fi or Cellular Data, to receive the push notifications.

Appendix A: Acronyms and Abbreviations

GUI	Graphical User Interface
RADIANT	RADar Intelligent Activity moNitoring

RADIAN - RADar-based Intelligent
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INTERFACE CONTROL DOCUMENT

INTERFACE CONTROL DOCUMENT FOR RADIAN

PREPARED BY:

Author Date

APPROVED BY:

Matthew Blackwell Date

Prof. Jang Date

Roman Venegas Date

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Figure 1: Electrical Power System

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1. Overview

This Interface Control Document (ICD) will cover the inputs, outputs, controls, and connections between the Physical, Data, and Wireless subsystems of the RADIAN system. This document will be continuously reviewed and updated as the RADIAN group continues to build, test, and integrate each subsystem. The ICD will include descriptions of each component being used in the system, how they interact with one another, and what data is output from each. The electrical connections and wireless interfaces, and protocols will be described in detail to ensure correct integration for success as described in the Concept of Operations (ConOps).

2. References and Definitions

2.1. References

IEEE 802.11

IEEE Standard for Information Technology – Telecommunications and Information Exchange between Systems

Jan. 23, 1996

2.2. Definitions

3GPP	3rd Generation Partnership Project
API	Application Programming Interface
A	Ampere
AI	Artificial Intelligence
CPU	Central Processing Unit
ConOps	Concept of Operations
g	Grams
GUI	Graphical User Interface
FSR	Functional System Requirements
HTTP	Hypertext Transfer Protocol
ICD	Interface Control Document
IEEE	Institute of Electrical and Electronics Engineers
ML	Machine Learning
mm	Millimeter
REST	Representational State Transfer

3. Physical Interface

3.1. Weight

The total weight of the RADIAN system, not including the Android device, will be less than or equal to 0.8 kilograms. This weight requirement enables the system to be properly mounted on the chosen room's wall without requiring heavy-duty hardware and support. Given that each subsystem has only a few components, the weight should not exceed the RADIAN group's given design.

3.2. Dimensions

3.2.1. Dimensions of Data Subsystem

Component	Length	Width	Height	Weight
Raspberry Pi 3	85.6 mm	56.5 mm	17 mm	41.2 g

Table 1: Data Subsystem Dimensions

3.2.2. Dimensions of Physical Subsystem

Component	Length	Width	Height	Weight
3.5Ah Power Board	100 mm	70 mm	15 mm	<90 g
Radome	190 mm	120 mm	55 mm	<230 g

Table 2: Physical Subsystem Dimensions

3.2.3. Dimensions of Wireless Subsystem

Component	Length	Width	Height	Weight
TI AWR6843ISK	103 mm	57 mm	16 mm	91 g

Table 3: Wireless Subsystem Dimensions

3.3. Mounting

3.3.1. Height

The RADIAN system will need to be mounted at a minimum of 2 meters off the ground. This allows the system to properly scan the room, have good line-of-sight, and detect if a fall has occurred.

3.3.2. Angle

The RADIAN system must be mounted to the wall at a 45-degree angle to allow for the maximum amount of area to be scanned by the radar. Having the system parallel to the

ground may prevent certain areas of the room from being scanned and, therefore, prevent proper detection.

3.3.3. Locations

The RADIAN system can be located in any common room of a home or building, including but not limited to living rooms, dining rooms, bedrooms, and kitchens. The RADIAN system is not rated for humidity or water exposure, so it cannot be used in the bathroom.

4. Thermal Interface

With the Raspberry Pi 3 and TI AWR6843 chip being enclosed in a 3D printed structure, the added thermal heat could cause reliability issues as the RADIAN system runs. Because the system runs continuously, this further increases the chance for potential failures due to overheating. To mitigate these risks, a small heatsink will be added to the Raspberry Pi 3 to ensure that it remains cool as it handles the data and accurately predicts if a fall occurred. A small 30 mm fan will also be installed to provide airflow across both the Raspberry Pi 3 and the AWR6843 chip. These solutions will allow the RADIAN system to run reliably and efficiently.

5. Electrical Interface

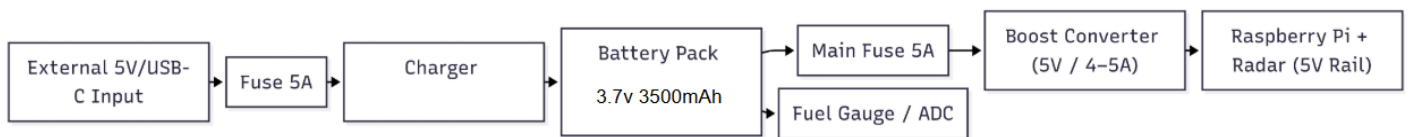


Figure 1: Electrical Power System

5.1. Primary Input Power

5.1.1 Electrical

The primary input power of the system will be two 18650 Li-ion cells in parallel, 3.7 V nominal each, for a 7 Ah total capacity. The battery voltage will be stepped up to 5V using a boost converter with a maximum continuous current of 5 A.

5.1.2 Connector

4-pin locking connector with two positive terminals and two negative terminals for the Raspberry Pi 3 and TI AWR6843.

Pinout (example): 1 = BAT+, 2 = BAT+, 3 = GND, 4 = GND.

5.1.3 Wiring

18–20 AWG wiring between the power board and the Raspberry Pi 3/ TI AWR6843. (Target $\leq 2\%$ drop on the 5 V rail.)

5.2. *Charging Interface*

5.2.1 Input

The battery system will be charged using a 5V, 1A wall adapter barrel jack. Charging at a low current will keep the batteries from overheating and ensure safety.

5.2.3 Connectors

The battery system will receive its power through a 5V barrel jack connected to a wall adapter.

5.3. *Grounding and ESD*

5.3.1 Ground

Single ground plane located on the power PCB.

5.3.2 ESD

TVS diodes on the barrel jack and on external connectors.

5.4. *User Control Interface*

The RADIAN system will come with a custom-designed phone application where the caregiver will be able to receive push notifications that a fall has been detected. The app may include a simple battery monitoring system to alert the user when the batteries need to be recharged.

6. Communications / Device Interface Protocols

6.1. *Wi-Fi Communication*

The Raspberry Pi 3 will use the IEEE 802.11b/g/n Wi-Fi standards through its onboard communication chip. These standards apply to Wi-Fi 1, Wi-Fi 3, and Wi-Fi 4, each offering new advancements in performance.

6.2. *Cellular Data*

The Gotify server will communicate with the phone application by 5G cellular data, given by the 3GPP standards for 5G wireless communication.

6.3. *Internet Protocols*

The Raspberry Pi 3 will post the alert message through the standard HTTP REST API protocol, allowing the message to be communicated from the Raspberry Pi 3 to the Gotify server.

6.4. *UART Serial Interface*

A UART serial interface will connect the Raspberry Pi 3 to the TI AWR6843ISK. Data shall be exchanged using packets containing radar object data. The Raspberry Pi shall use these packets and convert the radar output into the AI algorithm, which is then sent to the Gotify server.

7. Mobile Application Interface

7.1 *Communication Flow*

The Raspberry Pi 3 shall send event data to the Gotify server using an HTTP POST request. The Gotify server will then relay the data to the mobile app via WebSocket connection to receive messages in real time.

7.2 *User Interface*

The app shall display incoming alerts and system messages, battery life, and previous alerts if any in a custom-designed user interface.

7.3 *Platform Compatibility*

The app will be developed using Flutter for the user interface and Dart for the programming language. This gives the ability for the app to be run on Android devices.

7.4 *Notification Delivery*

The app shall generate a push notification when it has received an alert message from the Gotify server. The notifications will be displayed using the notification system of the Android device, to ensure the notifications are visible when the app is not open.

7.5 *Performance Requirements*

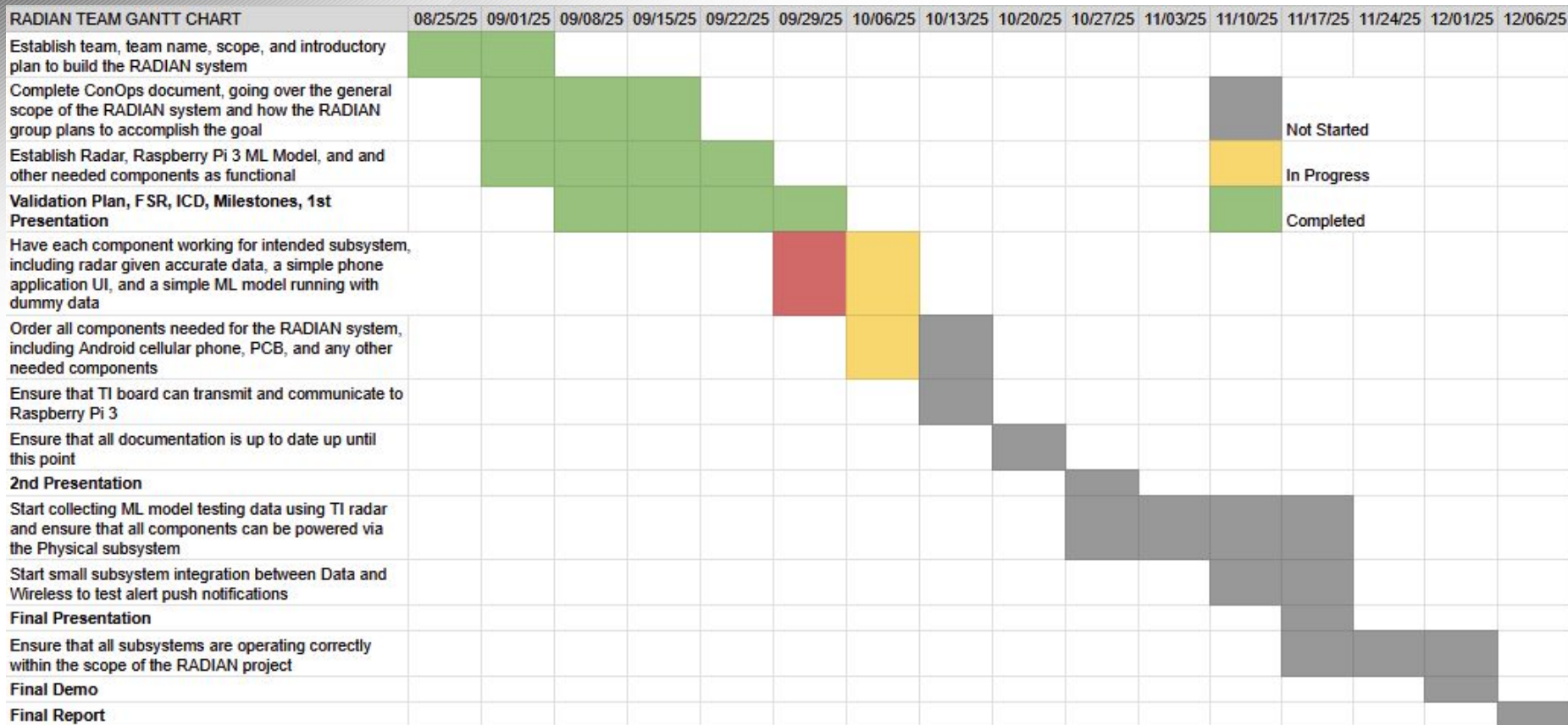
The alerts shall be delivered to the app within 5 seconds of the Raspberry Pi sending the alert to the Gotify server, under normal network conditions.



Integration/Execution Timeline



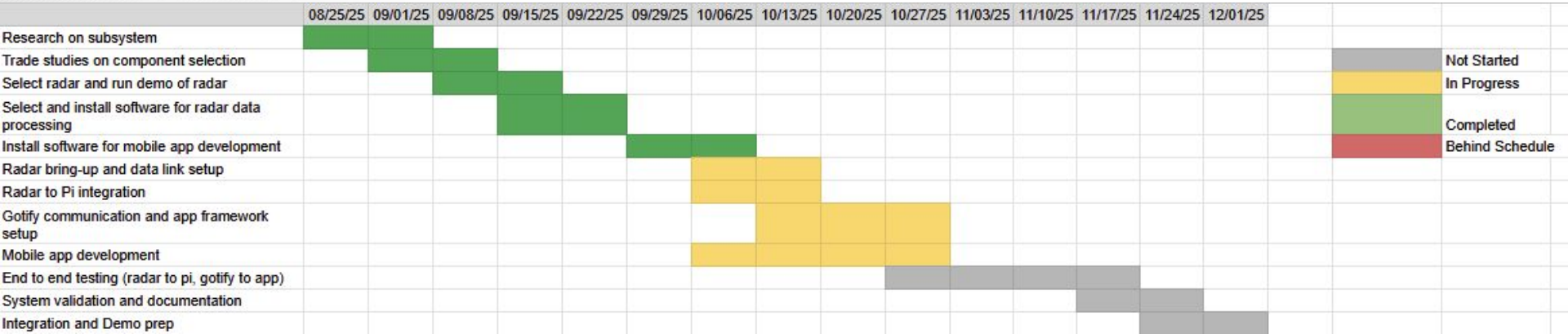
RADIAN Gantt Chart





Wireless Gantt Chart

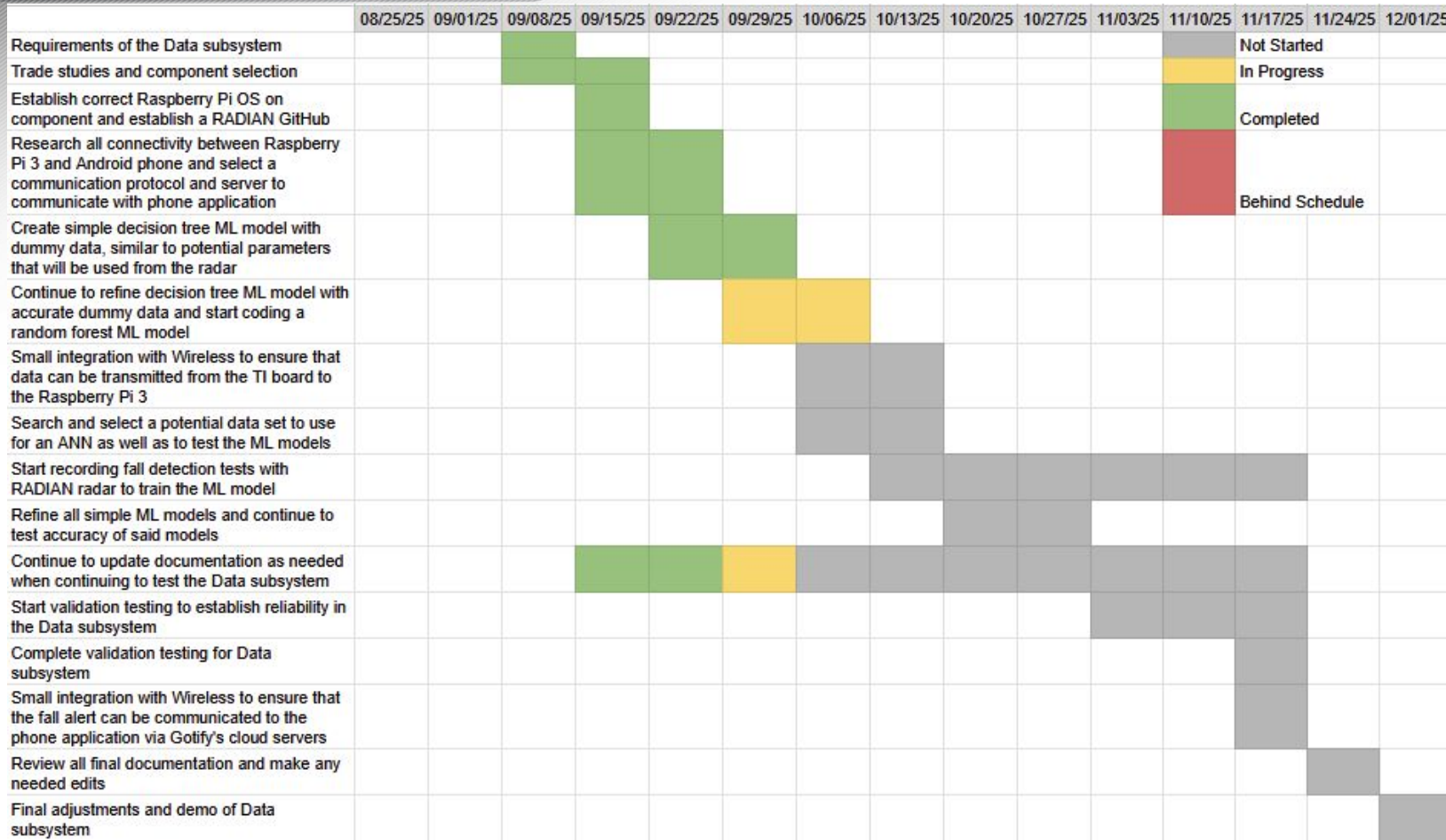
Mitchell Hoffman





Data Gantt Chart

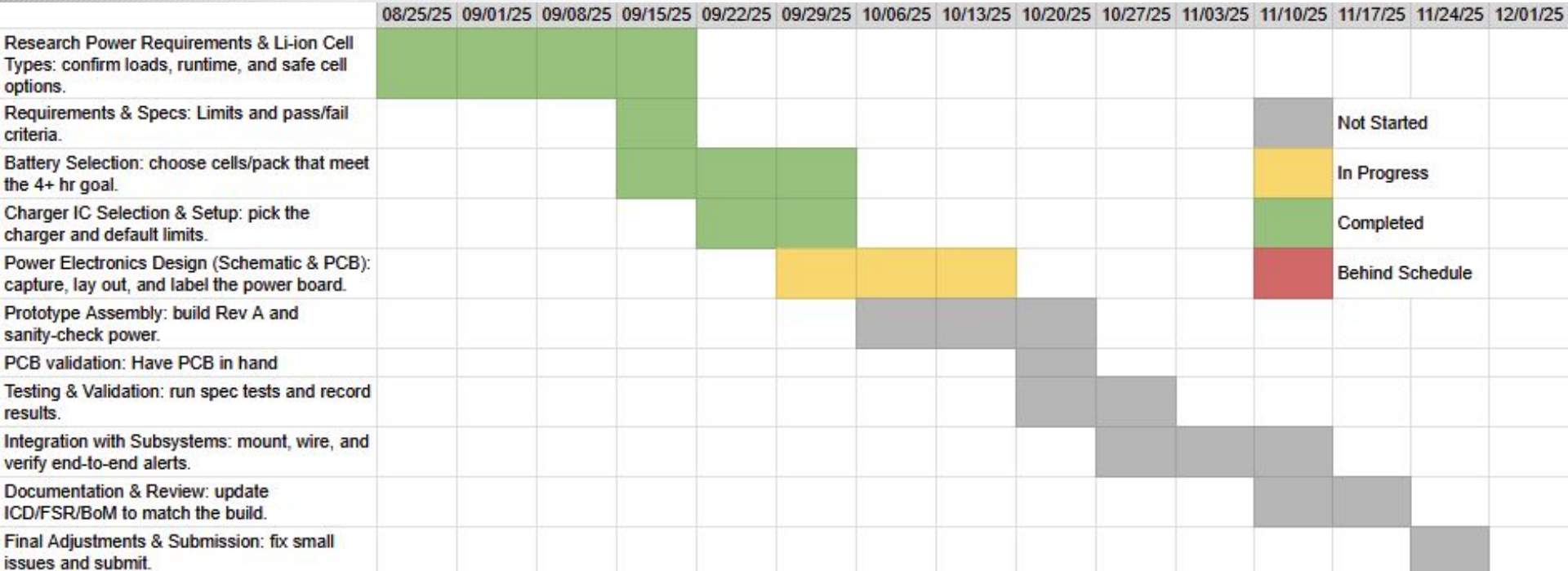
Matthew Blackwell





Physical Gantt Chart

Wyatt Johnson





Validation Plan

Paragraph #	Test Name	Success Criteria	Reasoning	Methodology	Status	Responsible Engineer
3.2.1.1.	Fall Detection Accuracy	The chosen ML model must report falls within an accuracy range of a minimum of 90%	If the machine learning model fails to report a fall, the high-risk individual will be at a higher risk of injury. If the machine learning model reports as false fall, the RADIAN system is not accurate and thus, may not be trusted by the user and the high-risk individual	The RADIAN Data subteam lead will continuously check accuracy of chosen ML model using Python code and visual confirmation.	Not Completed	Matthew
3.2.1.2.	Detection latency	The Raspberry Pi 3 is able to communicate with the RADIAN application within ~5 seconds of detected fall.	Quick response times are critical for alerting the caregiver if the high-risk individual fell so the appropriate actions can be taken.	The RADIAN group will send positive fall alert from the Raspberry Pi 3 to the Gotify server. The server will then send the alert to the RADIAN application to alert the caregiver. This process will be timed using the internal clock of the Raspberry Pi 3 as well as a physical stopwatch to ensure the notification was sent with ~5 seconds of the fall.	Not Completed	Matthew & Mitch
3.2.1.3.	Battery Life	The RADIAN system must operate for 4 hours with normal operation (People going in and out of a room, and two falls detected).	This amount of time will allow the system to stay function while the second battery is charging.	Physical team measures runtime on battery at average load with two scripted fall detection, pass if ≥ 4 h	Not Completed	Wyatt
3.2.1.4.	Communication Reliability	The RADIAN application must work within a ~200 meter radius from the system and will send an alert with a minimum of 95% success rate.	If the caregiver is not present in the home/building, the range must extend far enough to ensure the caregiver receives the notification	The RADIAN group will have on member stand outside of the ~200 meter radius and inside the ~200 meter radius to test if a push notification was received from the Raspberry Pi 3. There will also be a certain number of scripted notifications and log the amount of received notifications.	Not Completed	Matthew & Mitch
3.2.1.5.	Local Alert Functionality	Audible alarm at 70 dB at 1m when fall detected.	Having an alarm loud enough is essential for nearby individuals to hear the event.	Use SPL meter at 1m, trigger alert, record dB.	Not Completed	Wyatt
3.2.1.6.	Autonomous Operation	The RADIAN system shall operate autonomously with minimal user interaction.	Autonomy ensure there will be continuous monitoring and timely alerts.	Trigger a fall event and verify the app gets the notification and then trigger another fall event 1 hour after the first one was detected.	Not Completed	Mitch
3.2.2.1.	Radar Scan Area	Radar module effectively monitors a 20-30 square meters when mounted at a 45 degree downward angle.	Confirms the radar's field of view and detection performance align with the intended operational environment.	Mount radar module at a 45 degree downward tilt in a controlled 25 square meter test area. Place reflective targets or human subjects at multiple locations within test area. record detection coverage.	Not Completed	Mitch
3.2.2.2.	Mass	RADIAN system must be less than 0.8 kilograms.	A light system will be easily mountable on walls without heavy-duty hardware and support.	Weigh system with a scale once final design is completed.	Not Completed	Wyatt
3.2.2.3.	Volume Envelope	RADIAN system must be less than or equal to 5 in x 8 in x 14 in.	A Compact system will be easily mountable on walls allowing for relocation and light spaces.	Measure radome using tape measure.	Not Completed	Wyatt
3.2.3.1.1	Power Consumption	The peak power draw of the system shall not exceed 8 watts.	Lower power draw will allow the system to run longer.	Run worst case (Pi CPU load + radar high duty) and measure 5V current and calculate $P = VI$.	Not Completed	Wyatt
3.2.3.1.2	Input Voltage Level	System operate at 5.0V.	This is the required voltage for the Raspberry Pi 3 and TI AWR6843ISK to operate.	Apply 5.0V supply and measure the voltage at Pi and radar connectors.	Not Completed	RADIAN Group
3.2.3.2.1	Data Output	App interface accepts/decodes fall events from Gotify Cloud server.	This will allow for the caregiver to be notified when a fall is detected.	Send test events over REST/WebSocket and confirm app renders a notification.	Not Completed	Matthew & Mitch
3.2.3.2.3	Wireless Communication Output	Raspberry Pi 3 is able to communicate to the RADIAN application at an accuracy rate of 98%.	The alert system must work accurately in order for caregiver to know if the high-risk individual fell.	The RADIAN team will send a variety of notifications to the RADIAN application, both falls and not falls, and record the accuracy of each event occurring on the RADIAN application	Not Completed	Matthew & Mitch
3.2.4.1	Thermal	The RADIAN system will be capable of operating between 50 °F and 80 °F	The average temperature of many homes and buildings is between 50 °F and 80 °F, so the system must accommodate these varying temperatures	Room/chamber test from 50 °F and 80 °F, log temps and performance.	Not Completed	RADIAN Group
4.1	Wi-Fi Connection	The Raspberry Pi 3 connects to a specified Wi-Fi network and transmits a REST POST notification to the Gotify cloud server.	Validates that the RADIAN system can communicate with the Gotify server through Wi-Fi, meeting the requirement for wireless notification capability.	Monitor network status using Pi's terminal or a network monitor. Confirm a successful REST POST and verify message receipt on the Gotify server.	Not Completed	Matthew & Mitch
4.2	Wi-Fi or Cellular Data Connection	The caregiver's smartphone receives the RADIAN push notification within 5 seconds of it being sent by the Gotify server.	Confirms that the RADIAN notification system functions properly.	Trigger a test alert from the RADIAN device. Observe the phone's receipt of the notification and record delivery times.	Not Completed	Matthew & Mitch